

Phase 3 Inventory Summary for LinkedIn post creation

Inventory Phase 3 — Brief Summary

Phase 3 builds the **core inventory engine** using two-table accounting: `stock_ledger` (history) and `stock_reservations` (ownership).

1. Stock Receipt (Inbound Stock)

- `recordStockIn()`
- Adds material into inventory
- Writes only to **stock_ledger**
- Creates a STOCK_IN entry (+ quantity)

Effect: increases total inventory

2. Stock Reservation (Order Allocation)

- `reserveStock()`
- Checks available stock from ledger
- If sufficient → reserves stock for an order

Writes:

a. `stock_reservations`

- Marks ownership as ACTIVE

b. `stock_ledger`

- Adds RESERVED entry (negative quantity)

Effect: reduces available stock immediately

3. Stock Availability Logic

- Calculated dynamically from ledger:

Available = SUM(STOCK_IN + RESERVED)

- CONSUMED is ignored (no double deduction)
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4. Threshold Check (Warning System)

- checkThresholds()
- Runs after reservation
- Checks if available stock < material threshold
- Returns LowStockWarning list (non-blocking)

Effect: alerts only, does not stop orders

5. Stock Consumption (Dispatch Stage)

- consumeReservations()
- Runs during dispatch
- Bulk updates reservations:

ACTIVE → CONSUMED

- Writes CONSUMED entries in ledger (audit only)

Effect: finalizes usage, no change to availability

Core Design Model

stock_reservations → who owns stock (current state)

stock_ledger → full inventory history (source of truth)

Key Rule

- Reservation reduces availability
 - Consumption only records usage
 - Ledger is the single source for stock calculation
 - No double deduction occurs at dispatch
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Outcome of Phase 3

You now have a complete **inventory accounting system** that supports:

- Stock in tracking
- Order-based reservations

- Real-time availability calculation
- Threshold warnings
- Dispatch consumption audit

All without storing “available stock” directly anywhere.